RAUDIT / RCERT Boundary: Audit Transcripts and Certification Actions

Gio Della-Libera

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Abstract

RAUDIT and RCERT are reserved civic-evidence layers above the RCOUNT count ledger. RAUDIT names reusable audit, recount, and risk-limiting-audit transcript bundles; RCERT names certification actions, signoffs, evidence matrices, and scope caveats. Neither layer is currently a standalone implementation boundary: RAUDIT remains embedded in RCOUNT through the V-series audit-algorithm replay contracts, and RCERT is not built. This paper records the reservation boundary. Audit methods may be replayed from public evidence, and official actions may be recorded against verified evidence, but neither layer owns the base count ledger or silently changes RCOUNT verifier results.

1 Introduction

The count ledger is the center of the RCOUNT package. It carries the declared election totals, CVRs, manifests, source hashes, reconciliation transcripts, and verifier results that answer one narrow question: do the public count records reconcile under the declared package rules?

Two other kinds of public evidence sit naturally above that ledger. Audit and recount methods produce transcripts: assertions, sample observations, risk limits, replay statistics, stopping decisions, and boundary notes. Certification processes produce official actions: board minutes, motions, signoffs, dates, evidence matrices, and legal caveats. Both are reusable enough to deserve names. The reserved names are RAUDIT for audit transcript bundles and RCERT for certification actions.

The reservation is intentionally conservative. Pulling audit transcripts out of RCOUNT too early would fragment the verifier before there is a second production consumer for the same transcript records. Building RCERT before real public certification source artifacts are identified would model a legal process in the abstract instead of binding actions to evidence. For now, RAUDIT remains embedded in RCOUNT as the V-series audit-algorithm replay contracts, and RCERT remains an unbuilt boundary.

This paper records that boundary. It does not introduce a standalone RAUDIT crate or a standalone RCERT crate. It defines what each layer would own, what it must read, and what it must not change.

2 Ownership Contract

RAUDIT is the reserved owner of reusable audit, recount, and RLA transcript bundles when those transcripts need to stand outside a full RCOUNT package. The current embedded form is the V-series audit replay surface: BRAVO, Minerva, Kaplan-Markov, ALPHA, SHANGRLA, stratified audit, batch comparison, RAIRE, Bayesian, and SOBA contracts each preserve method evidence and report whether the method can be replayed, fails, or remains at a boundary.

RCERT is the reserved owner of certification actions. It would record what an official body did, when it did it, who signed or voted, what evidence was cited, and what legal or administrative caveats limit the action. RCERT reads count, audit, log, and custody evidence by reference. It does not recompute count arithmetic as an independent source of truth.

Layer	Owns	Must not do
RCOUNT	Base count ledger, source hashes, reconciliation transcripts, and count verifier results.	Transfer ledger ownership to downstream audit or certification records.
RAUDIT	Reusable method transcripts: method ids, assertions, samples, risk limits, replay statistics, stopping decisions, and boundary fields.	Mutate RCOUNT ledgers, certify an election by itself, or hide method assumptions in prose.
RCERT	Official actions, signoffs, minutes, evidence matrices, dates, signatories, and scope caveats.	Recompute count arithmetic without binding to RCOUNT, or turn audit or statistical records into official findings by implication.
RLOG/RCHAIN	Event logs, machine or administrative status records, seals, transfers, observers, and custody caveats.	Replace count verification or certification action records.

The resulting dependency direction is simple. RAUDIT may read RCOUNT, RCTX, and optionally RHIST when a method depends on units, strata, or geography. RCERT may read RCOUNT, RAUDIT, RLOG, and RCHAIN. Neither downstream layer writes the base count ledger, and neither layer is allowed to silently revise an RCOUNT verifier result.

3 Deferral and Boundary

The downstream evidence boundary specification reserves RAUDIT and RCERT but does not implement them. The civic-evidence access-pattern specification gives the read direction, and the RCOUNT audit-algorithm roadmap shows why the current audit work still belongs inside RCOUNT [Della-Libera, 2026a]. The V-series contracts are still proving the common transcript shape: method ids, assertion ids, sample or batch references, risk limits, per-step statistics, final decisions, and explicit boundary outcomes.

RAUDIT should therefore stay embedded until a second production consumer needs the same audit transcript independently of an RCOUNT package. That consumer might be a public case bundle, a cross-jurisdiction audit library, or a certification record that wants to cite a transcript package by hash. Until that pressure exists, extracting RAUDIT would create another package boundary without making the evidence more verifiable.

RCERT has a different deferral reason. Certification is source-artifact driven. Before schema work begins, the project needs real public certification artifacts: board minutes, canvass reports, motions, official certificates, signatory records, evidence checklists, legal caveats, and any referenced audit, log, or custody exhibits. Without those artifacts, RCERT would risk inventing a generic workflow rather than preserving what officials actually publish.

One decision remains open: whether the V-series audit-algorithm transcripts need standalone RAUDIT packages at all. Some methods may remain most useful as embedded RCOUNT replay contracts. Others may become reusable exhibits that a certification or public-case layer should reference directly. The boundary rule does not answer that packaging question. It only fixes ownership: RAUDIT can replay method evidence. RCERT can record official actions. Neither owns the base count ledger, and neither should silently change RCOUNT verifier results.

4 Conclusion

RAUDIT and RCERT are useful names, but they are not yet standalone implementation layers. RAUDIT is reserved for reusable audit, recount, and RLA transcripts; today those transcripts remain embedded in RCOUNT through the V-series replay contracts. RCERT is reserved for certification actions, signoffs, evidence matrices, and scope caveats; it should wait until public certification source artifacts are identified.

The boundary protects the verifier. RCOUNT continues to own the base count ledger and count-verification result. Future RAUDIT records may replay method evidence, and future RCERT records may cite that evidence in official actions, but neither layer may rewrite count facts or turn a downstream action into a silent change in verifier outcome.

References

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